SO cross-cl pipeline: decision tree v3 — Bayesian search + profile scheme

left: search for moments with the evidence (as in v2) · right: new profile scheme, replacing the frequentist branch

A · Bayesian search for moments (evidence, r free)

run BBPower without moments → $M_{0,r}$

is χ^2 good? (PTE > 0.01)
- No → Res_wr: residuals of $M_{0,r}$ all fixed at its best fit (r too); test d and s, each with its own β free
- Yes → is r compatible with zero at 1σ?
  - No → Res_wr: residuals of $M_{0,r}$ all fixed at its best fit (r too); test d and s, each with its own β free
  - Yes → upper limit on r ($M_{0,r}$)

Res_wr: moment detected? (0 outside HPD 68%)
- Yes → $M_{1,r}$ = $M_{0,r}$ + best moment (residual evidence), all parameters free
  - is $M_{1,r}$ preferred over $M_{0,r}$? (ln B > 0)
    - Yes → $M_{0,r}$ = $M_{1,r}$ (pruned), start again
    - No (*) → Res_r0: residuals of $M_{0,0}$ (r = 0) all fixed, own β free; test d, s, dT, d2, s2, ds1, ds2
- No → Res_r0: residuals of $M_{0,0}$ (r = 0) all fixed, own β free; test d, s, dT, d2, s2, ds1, ds2

Res_r0: moment detected? (HPD 68%)
- Yes → best one (residual evidence): $M_{1,r}$, all parameters free
  - is $M_{1,r}$ preferred over $M_{0,r}$? (ln B > 0)
    - Yes → $M_{0,r}$ = $M_{1,r}$ (pruned), start again
    - No → $M_{1,0}$: same model, r = 0
      - is $M_{1,0}$ preferred over $M_{0,r}$? (ln B > 0)
        - Yes → $M_{0,r}$ = $M_{1,0}$ (pruned), start again
        - No → back to step 0 (orange case)
- No → back to step 0

(*) if $M_{0,r}$ has no moments yet, the tree stops after Res_wr: → B with C = the moment just tried (orange case)

B · Profile scheme (new): uses r as a profile, not marginalised

M = current model, C = candidate moment · PolyChord only where an evidence is needed (steps 2, 3) · profiles: max likelihood · parameters: emcee

0 — starting point

- $M_{1,0}$ preferred → M = $M_{1,r}$, the best model found: no candidate, go straight to step 3
- a moment was tried as $M_{1,r}$ and lost ($M_{1,0}$ not preferred) → M = $M_{0,r}$, candidate C = that moment
- nothing new detected in Res_r0 → M = $M_{0,r}$, no candidate: go straight to step 4 (r from its profile)

1 — profile of r of M + C

r fixed on a grid, all other parameters re-fitted (max likelihood) → r* = minimum

2 — is M + C preferred over M?

each model at its own best fit (M + C at r*, M at the minimum of its profile); free only the moment spectra + the same β in both models; ln B > 0
- Yes → M = M + C (C enters the model)
- No → C rejected (not proposed again)

3 — Res_wr*: residuals of M re-fitted at r = r*

all groups not yet in the model and not rejected (also dT, d2, s2, ds1, ds2): screening with emcee (0 outside HPD 68%), then evidence against the null (same point, moment amplitudes = 0, only β free): passes if ln B > 0

does a new moment pass?
- Yes: C = the best one, back to 1
- No → **4**

4 — result: r from the profile of M used as a likelihood × prior of the tree → median, 68%, σ, r_95 (also with r ≥ 0); other parameters: emcee of M with r = r* fixed

0 within 3σ of r?
- Yes → upper limit
- No → r detection